MAT 110 Differential Calculus & Coordinate Geometry

Week 1 Topic: → Limits

→ Continuous and Discontinuous function

→ Rational Function and Asymptotes

→ Computing Limits

- Algebric Manipulation
- Squeezing or Squeeze Theorem
- Change of Variable
- L' Hôspital' Rule

Limit → What is Limit?

A limit looks at what happens to a function when the input approaches a certain value. Example $\lim_{x \to a} f(x)$. It denotes the limit of $f(x)$ as 'x' approaches 'a'.

Consider $f(x) = \frac{4}{3}x - 4$

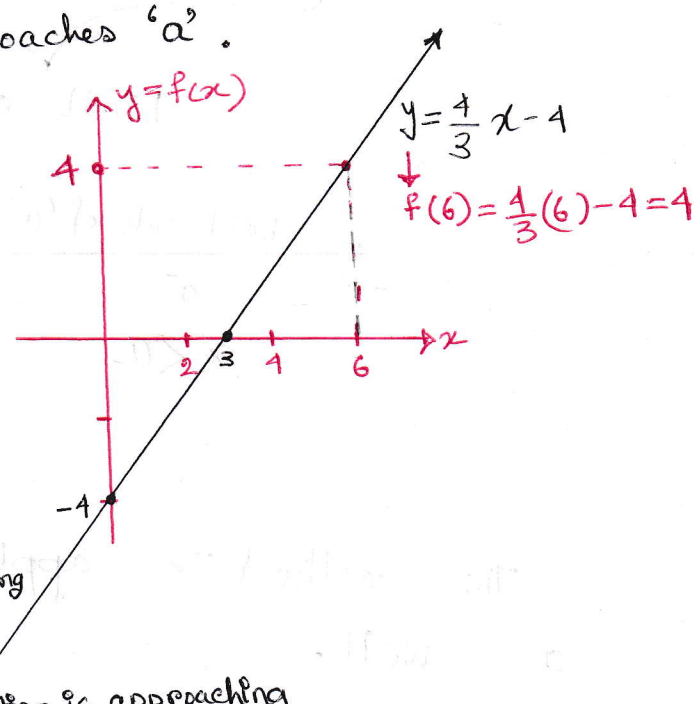
y → output

x → input y = f(x)

if $x = 0 \Rightarrow f(x) = -4$

if $y = 0 \Rightarrow 0 = \frac{4}{3}x - 4$

$x = 3$



Limit Notation

The "lim" tells us we are looking for a limit value, not a function value

This tells us which function we are working with

$$\lim_{x \to 6} f(x) = 4$$

This tells us what the variable is, and what it is approaching

This this the value the function is approaching

Existence of Limit

Limit of a function exist if L.H.L = R.H.L

Or, Limit exists if $\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x)$

$$f(x) = \begin{cases} p(x) ; & x \geq a \\ q(x) ; & x < a \end{cases}$$

If we want to know $\lim_{x \to a} f(x) = ?$

It depends on two cases:

$a = 4$

$x < 4$ | $x > 4$

$\lim_{x \to 4^-}$ | $\boxed{x = 4}$ | $\lim_{x \to 4^+}$

$a = -5$

$x < -5$ | $x > -5$

$\lim_{x \to -5^-}$ | $\boxed{x = -5}$ | $\lim_{x \to -5^+}$

$$\boxed{\lim_{x \to a} f(x) = ?}$$

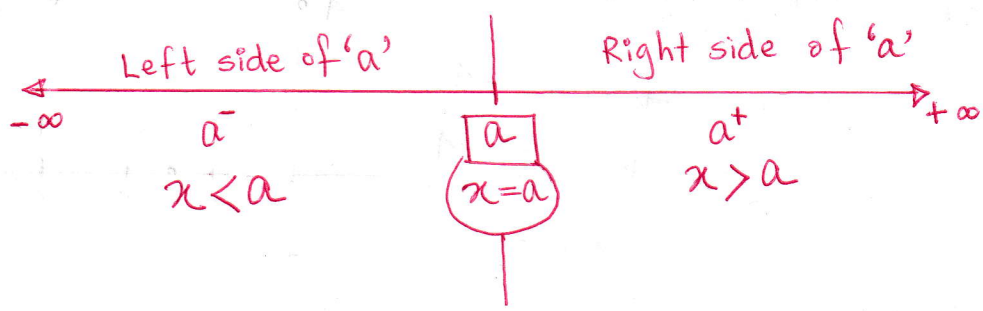
$$\boxed{\lim_{x \to a^-} f(x)} \rightarrow p(x)$$

Left hand limit (L.H.L)

$$\boxed{\lim_{x \to a^+} f(x)} \rightarrow q(x)$$

Right hand limit (R.H.L)

Limit exists if these two are equal.



This method is applicable for Continuous function as well.

Continuous Function:

A function $f(x)$ is said to be continuous if

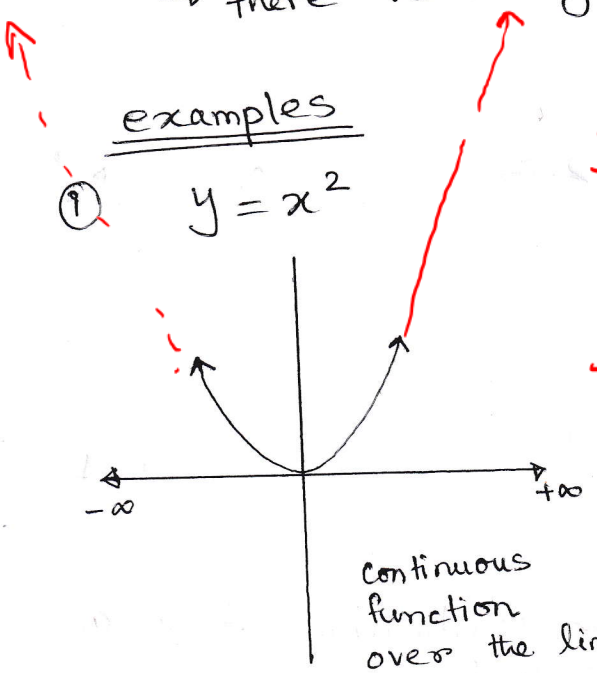
→ there is no discontinuity

OR

→ there is no gap in the graph

examples

(i) $y = x^2$



continuous function over the limit $(-\infty, +\infty)$

$\frac{n}{0} = \infty$

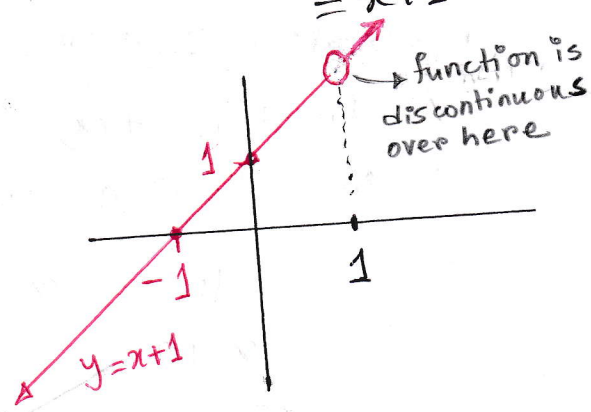
$\frac{0}{n} = 0$

(ii) $y = \frac{x^2-1}{x-1}$

$x \neq 1$ otherwise y will be undefined

Now $y = \frac{x^2-1}{x-1} = \frac{(x+1)\cancel{(x-1)}}{\cancel{(x-1)}}$

$= x+1$



Several Examples of Discontinuous functions:

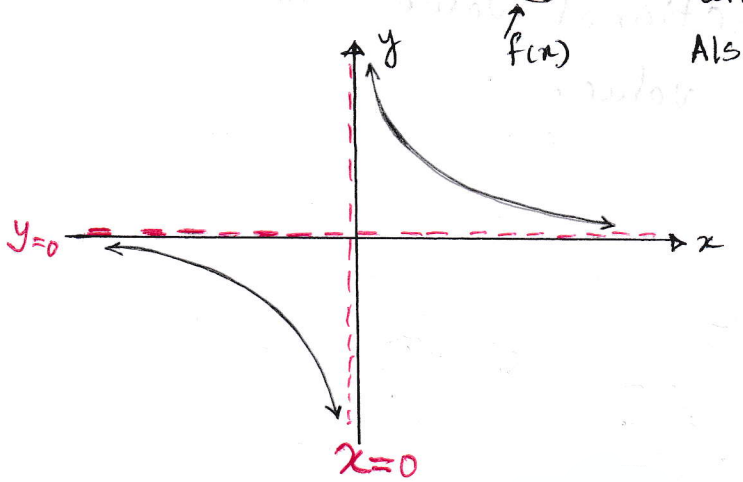
(a) $f(x) = \frac{1}{x}$ or $y = \frac{1}{x}$ → $x \neq 0$ otherwise function will be undefined.

↳ $f(x)$

Also $y = \frac{1}{x} \Rightarrow xy = 1 \Rightarrow x = \frac{1}{y}$

$f(y)$

$y \neq 0$ otherwise $f(y)$ will be undefined.



(b) $f(x) = \dfrac{x-4}{x+1}$

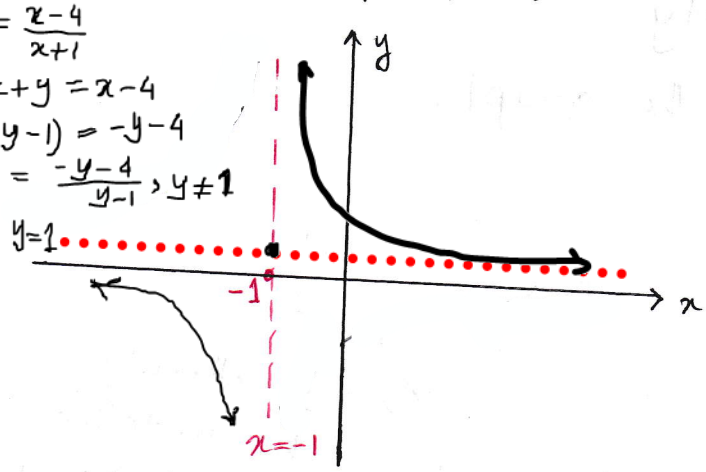
$x \neq -1$ o/w $f(x) = \infty$

$y = \dfrac{x-4}{x+1}$

$yx + y = x - 4$

$x(y-1) = -y-4$

$x = \dfrac{-y-4}{y-1}, \; y \neq 1$



Discontinuity at $x = -1$

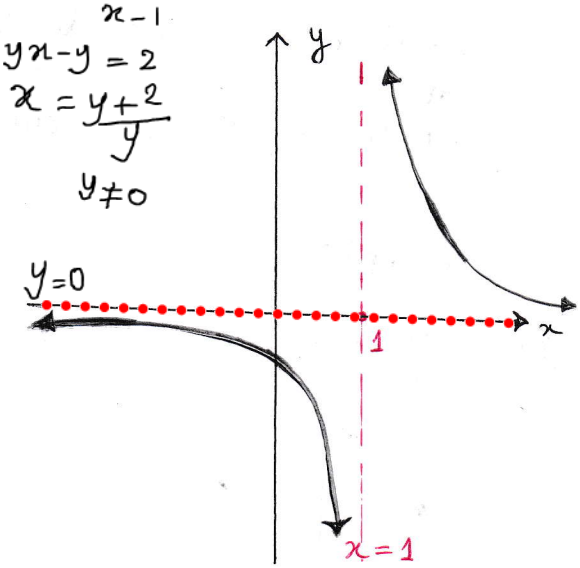
(c) $f(x) = \dfrac{2}{x-1}$; $x \neq 1$

$y = \dfrac{2}{x-1}$

$yx - y = 2$

$x = \dfrac{y+2}{y}$

$y \neq 0$



Discontinuity at $x = 1$

Mathematical Definition

A function $f(x)$ is said to be continuous at $x=a$ if the following condition is satisfied

$$\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = f(a)$$

Existance of Limit $= f(a)$

$\therefore$ Function is continuous if L.H.L = R.H.L = $f(a)$

So at $x=a$, the functional value should be equal to the limit value.

Rational Function:

Rational number: $\frac{p}{q}$, p & q both are integers and $q \neq 0$.

Rational function: $f(x) = \frac{p(x)}{q(x)} = \frac{\text{numerator}}{\text{denominator}}$

$= \frac{\text{constant or variable}}{\text{variable } (x\text{-type})}$

denominator must be independent variable type.

Examples

Rational Function:

(a) $f(x) = \frac{x-12}{4x^2+x+1}$

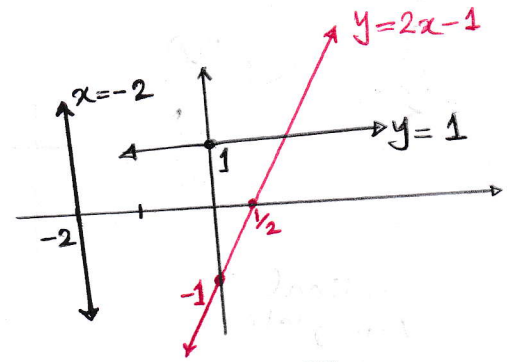
(b) $G(x) = \frac{8x^2-x+2}{4x^2-1}$

(c) $f(x) = \frac{2x^3+x^2-7x-3}{x^2-4}$

(d) $R(x) = \frac{1}{x^2-4}$

Not Rational Function:

(e) $R(x) = \frac{x+1}{2} = \frac{1}{2}x + \frac{1}{2}$



Asymptotes

Slope	Lines	Equation	Examples
$m = \infty$	Vertical ↕	$x = a$	$x=2$, $x=-4$, $x=0$, $x=\frac{1}{2}$ etc
$m = 0$	Horizontal ↔	$y = b$	$y=-3$, $y=9$, $y=0$, $y=-\frac{1}{3}$ etc
$m\ (\pm)$	Oblique	$y = mx+c$	$y=2x-1$

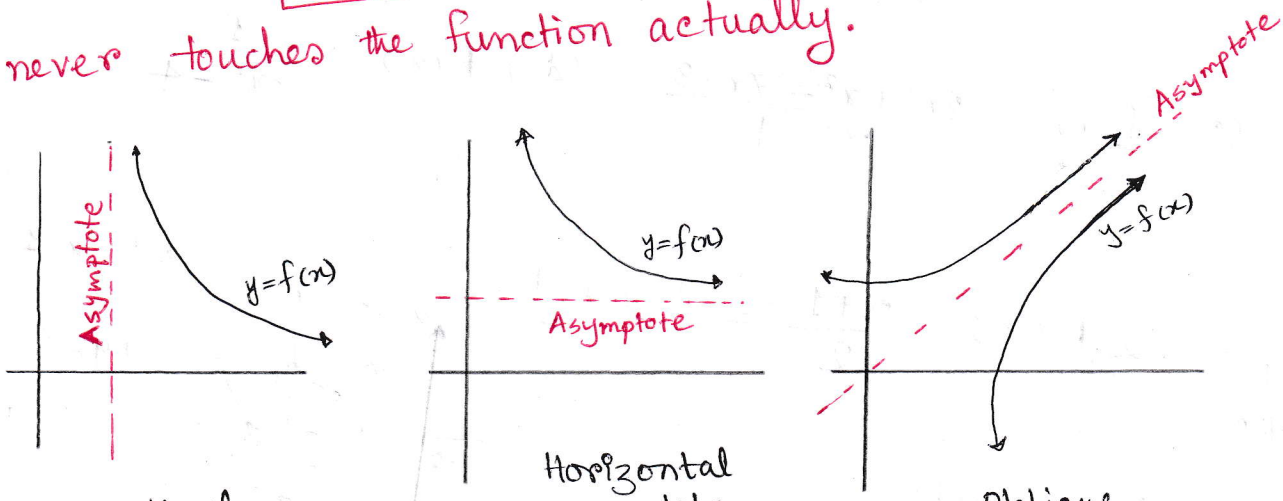
Classification of Asymptote

Asymptote

- Vertical Asymptote → $\boxed{x=a}$
- Horizontal Asymptote → $\boxed{y=b}$
- Oblique Asymptote → $\boxed{y=mx+c}$

Definition:

An asymptote is a straight line that seems to touch a function $\boxed{y=f(x)}$ at infinity $(+\infty\,/,-\infty)$, but never touches the function actually.



Vertical Asymptote

Horizontal Asymptote

Oblique Asymptote

We will need these TWO for MAT 110

Computing Limits:

Properties of Limit

Theorem: Let a be a real number, and suppose that

$$\lim_{x \to a} f(x) = L_1 \quad \text{and} \quad \lim_{x \to a} g(x) = L_2$$

That is, the limit exist and have values L_1 and L_2, respectively. Then:

(a) $\lim_{x \to a} [f(x) \pm g(x)] = \lim_{x \to a} f(x) \pm \lim_{x \to a} g(x) = L_1 \pm L_2$

Addition OR subtraction

(b) $\lim_{x \to a} [f(x)\, g(x)] = \left(\lim_{x \to a} f(x)\right)\left(\lim_{x \to a} g(x)\right) = L_1 L_2$

Multiplication

(c) $\lim_{x \to a} \dfrac{f(x)}{g(x)} = \dfrac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)} = \dfrac{L_1}{L_2}$, note that $L_2 \neq 0$

division

(e) $\lim_{x \to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \to a} f(x)} = \sqrt[n]{L_1}$; provided that $L_1 > 0$ if n is even

root

$(-2)^2 = +4$ $\quad (-2)^3 = -8$

$(+2)^2 = +4$ $\quad (+2)^3 = +8$

Ex: $\sqrt[4]{16} = 2$, $\sqrt[4]{-16} = \infty$, $\sqrt[3]{8} = 2$, $\sqrt[3]{-8} = -2$

These statement are also true for one-sided limits as $x \to a^-$ or as $x \to a^+$

Example:

$$\lim_{x \to 1} \left(x^2 + \frac{x^3+1}{x^2+1} + 2x \sin\left(\pi \sqrt{3x^2+1}\right)\right)$$

$$= 1^2 + \frac{1^3+1}{1^2+1} + 2(1) \sin\left(\pi \sqrt{3(1)^2+1}\right)$$

$$= 1 + 1 + 2\sin 2\pi = 2$$

Various Methods to Evaluate Limits:

$\boxed{1}$ Algebric Manipulation

$$\lim_{x \to 3} \frac{x^2-9}{x-3}$$

$$= \lim_{x \to 3} \frac{(x+3)(x-3)}{(x-3)}$$

$$= \lim_{x \to 3} x+3$$

$$= 3+3$$

$$= 6$$

$$\lim_{x \to -4} \frac{2x+8}{x^2+x-12}$$

$$= \lim_{x \to -4} \frac{2x+8}{x^2+4x-3x-12}$$

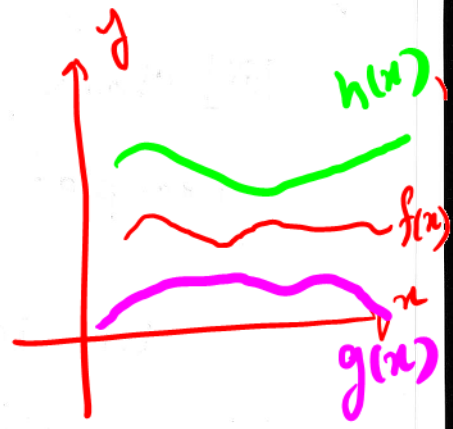
$$= \lim_{x \to -4} \frac{2x+8}{x(x+4)-3(x+4)}$$

$$= \lim_{x \to -4} \frac{2(x+4)}{(x+4)(x-3)}$$

$$= \lim_{x \to -4} \frac{2}{x-3}$$

$$= \frac{2}{-4-3} = \frac{-2}{7}$$

ii Squeeze Theorem

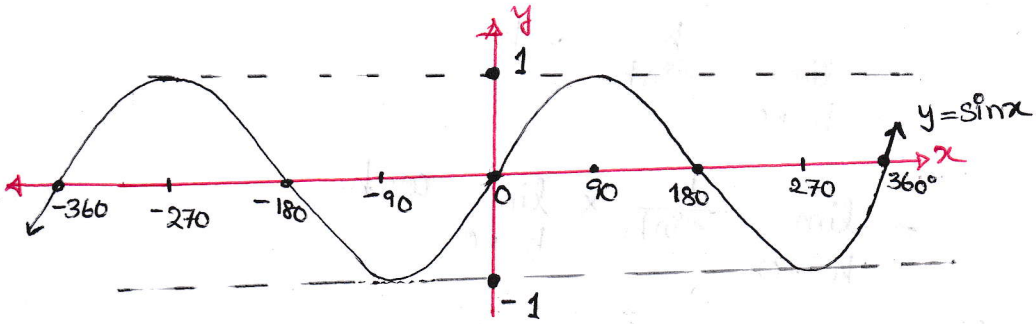


Suppose : $g(x) \le f(x) \le h(x)$

If $\lim_{x \to a} g(x) = L$ and $\lim_{x \to a} h(x) = L$

Then $\lim_{x \to a} f(x) = ? = L$

Example: $\lim_{x \to 0} x \sin \frac{1}{x} = ?$



We know, $-1 \le \sin x \le 1$

$\therefore -1 \le \sin\left(\frac{1}{x}\right) \le 1$

$\Rightarrow -1(x) \le (x) \sin\left(\frac{1}{x}\right) \le 1(x)$ { multiply by x in all sides to create the main (given) function

$\Rightarrow -x \le x \sin\left(\frac{1}{x}\right) \le x \longrightarrow g(x) \le f(x) \le h(x)$

comparing

$\lim_{x \to 0} (-x) = 0$

$\lim_{x \to 0} (x) = 0$

$\lim_{x \to 0} \left(x \sin \frac{1}{x}\right) = 0$ by Squeez Theorem.

iii L' Hôspital's Rule:

We need to know all the rules of differentiation

If we have to evaluate $\lim_{x \to a} \frac{f(x)}{g(x)}$ and

initially if we have $\lim_{x \to a} \frac{f(x)}{g(x)} = \underbrace{\frac{0}{0} \text{ or } \frac{\infty}{\infty}}$

This form is known as Indeterminate form

Note

(*) $\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$ if $\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{0}{0}$ or $\frac{\infty}{\infty}$

(*) If the limiting part becomes $\frac{0}{0}$ or $\frac{\infty}{\infty}$ again, then differentiation will be continued.

Example:

(a) $\lim_{x \to 1} \frac{\ln x}{x-1}$ $\quad \frac{\ln 1}{1-1} = \frac{0}{0}$ form

$= \lim_{x \to 1} \frac{\frac{d}{dx}(\ln x)}{\frac{d}{dx}(x-1)}$

$= \lim_{x \to 1} \frac{1/x}{1-0}$

$= \lim_{x \to 1} \frac{1}{x}$

$= \frac{1}{1} = 1$

(b) $\lim_{x \to 0} \frac{1}{x} - \frac{1}{\sin x}$ $\quad \frac{1}{0} - \frac{1}{\sin 0} = \infty - \infty = \infty$

$= \lim_{x \to 0} \frac{\sin x - x}{x \sin x}$ $\quad \frac{\sin 0 - 0}{0 \sin 0} = \frac{0-0}{0} = \frac{0}{0}$ form

$= \lim_{x \to 0} \frac{\frac{d}{dx}(\sin x - x)}{\frac{d}{dx}(x \sin x)}$

$= \lim_{x \to 0} \frac{\cos x - 1}{\sin x + x \cos x}$ $\quad \frac{\cos 0 - 1}{\sin 0 + 0 \cdot \cos 0} = \frac{1-1}{0+0} = \frac{0}{0}$ form again

$= \lim_{x \to 0} \frac{\frac{d}{dx}(\cos x - 1)}{\frac{d}{dx}(\sin x + x \cos x)}$

$= \lim_{x \to 0} \frac{-\sin x - 0}{\cos x + \cos x - x \sin x}$

$= \frac{0}{1+1-0} = \frac{0}{2} = 0.$

iv Change of variable

Example: $\lim_{x \to 0} \frac{\tan^{-1} x}{x} = ?$

Let $\tan^{-1} x = h$ introducing new variable 'h'

$\therefore x = \tan h$

when $x \to 0$ then $h \to 0$

Now $\lim_{x \to 0} \frac{\tan^{-1} x}{x} = \lim_{h \to 0} \frac{h}{\tan h}$

$$\frac{\frac{a}{b}}{\frac{c}{d}} = \frac{a}{b} \times \frac{d}{c}$$

$= \lim_{h \to 0} \frac{h}{\frac{\sin h}{\cos h}}$

$= \lim_{h \to 0} \frac{h}{\sin h} \cdot \cos h$

$= \lim_{h \to 0} \frac{h}{\sin h} \times \lim_{h \to 0} \cos h$

$= 1 \times 1$

$= 1$

$\lim_{h \to 0} \frac{h}{\sin h} = \frac{0}{0}$

if $\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{f(a)}{g(a)} = \begin{cases} \frac{0}{0} \text{ OR} \\ \frac{\infty}{\infty} \end{cases}$

then $\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$

L'Hospital's Rule

$= \lim_{h \to 0} \frac{1}{\cos h}$ (apply L'Hospital's rule)

$= \frac{1}{\cos(0)}$

$= \frac{1}{1}$

$= 1$